

Section I: Data Structures [15 Marks]

Q1) Create random array of n integers and sort the array in ascending order by using recursive Merge sort algorithm

Answer:

```
#include <stdio.h>
#include <stdlib.h>

void merge(int a[], int l, int m, int r)
{
    int b[100], i = l, j = m + 1, k = 0;

    while(i <= m && j <= r)
    {
        if(a[i] <= a[j])
            b[k++] = a[i++];
        else
            b[k++] = a[j++];
    }

    while(i <= m)
        b[k++] = a[i++];
    while(j <= r)
        b[k++] = a[j++];

    for(i = l, k = 0; i <= r; i++, k++)
        a[i] = b[k];
}

void mergesort(int a[], int l, int r)
{
    int m;

    if(l < r)
    {
        m = (l + r) / 2;
        mergesort(a, l, m);
        mergesort(a, m + 1, r);
        merge(a, l, m, r);
    }
}

int main()
{
    int a[100], n, i;
```

```

    printf("Enter n: ");
    scanf("%d", &n);

    if(n <= 0 || n > 100)
    {
        printf("n must be between 1 and 100");
        return 0;
    }

    for(i = 0; i < n; i++)
        a[i] = rand() % 100;

    printf("Original array: ");
    for(i = 0; i < n; i++)
        printf("%d ", a[i]);

    mergesort(a, 0, n - 1);

    printf("\nSorted array: ");
    for(i = 0; i < n; i++)
        printf("%d ", a[i]);

    return 0;
}

```

Section II: Database Management Systems [15 Marks]

Q 2) Create the following tables (primary keys are underlined).

Account (acctno ,acct_type, balance, branch_name)

Customer (custno, cust_name, cust_city)

Relationships: Customer-Account: One to Many.

Constraints: acct_type should be “saving” or “current” and balance is greater than zero.

Q 2(A)

A. Create the above database in PostGreSQL and insert sufficient records. [5 marks]

Answer:

```

CREATE TABLE Customer(
    custno INT PRIMARY KEY,
    cust_name VARCHAR(50),
    cust_city VARCHAR(50)
);

```

```
CREATE TABLE Account(
acctno INT PRIMARY KEY,
acct_type VARCHAR(10)
        CHECK(acct_type IN ('saving','current')),
balance NUMERIC(12,2) CHECK(balance > 0),
branch_name VARCHAR(50),
        custno INT REFERENCES Customer(custno)
ON DELETE CASCADE
);
```

```
INSERT INTO Customer VALUES
(1,'Ramnath','Mumbai'),
(2,'Sanjay','Nashik'), (3,'Meena','Pune');
```

```
INSERT INTO Account VALUES
(101,'saving',8000,'M.G.Road',1),
(102,'current',20000,'M.G.Road',2),
(103,'saving',3000,'Camp',3);
```

Q 2(B)

B. Execute the following queries in PostGreSQL (any 3) [6 marks]

i. Display information of all saving accounts having balance greater than 5000.

Answer:

```
SELECT * FROM Account
WHERE acct_type = 'saving' AND
balance > 5000;
```

ii. Count customers in city "Pune".

Answer:

```
SELECT COUNT(*) FROM Customer WHERE
cust_city = 'Pune';
```

iii. Find the total balance at branch "M.G.Road".

Answer:

```
SELECT SUM(balance) FROM Account WHERE
branch_name = 'M.G.Road';
```

iv. Delete the record whose cust_name is "Ramnath".

Q 2(C)

C. Write a query to find the number of account holders for each city. [4 marks]

Answer:

```
SELECT C.cust_city,  
       COUNT(DISTINCT C.custno) AS account_holders  
FROM Customer C  
JOIN Account A ON C.custno = A.custno  
GROUP BY C.cust_city;
```

Q 3) Viva [5 marks]